

Finding the Optimal Regularization Parameter in Distribution of Relaxation Times Analysis

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The distribution of relaxation times (DRT) method allows the direct interpretation of electrochemical impedance data, yielding an increased resolution in the frequency domain. Calculating the DRT from experimental impedance spectra, however, is an intrinsically ill-posed problem requiring special mathematical treatments such as, for example, the Tikhonov regularization. In this study, we propose a new approach for finding optimal regularization parameters in the Tikhonov regularization. The new test function is based on repetitive impedance measurements and a simple data resampling with a subsequent

variance investigation. Furthermore, our new approach is combined with already published procedures considering the determination of best regularization parameters. Here, the combination of several tests enables a clear assignment of an upper and a lower boundary for suitable regularization parameters. Finally, both tests were applied to simulated as well as to experimental impedance data for validation. To analyze the sensitivity of the approach, varying error extent was generated to the synthetic and experimental measurements.

1. Introduction

Driven by the enormous efforts in battery and fuel cell research and development, analytical methods for their in-depth characterization become increasingly important. One of the most powerful tools to investigate electrochemical systems is the electrochemical impedance spectroscopy (EIS). Its non-invasive nature and its ability to distinguish and identify different physicochemical processes based on their characteristic time constants make impedance spectroscopy an often used and outstanding technique.^[1] However, the interpretation of measured impedance data oftentimes requires a good knowledge about the investigated system's processes. Furthermore, when the time constants of individual processes are close to each other, data analysis can become complicated and ambiguous as the latter's signals overlap in frequency domain.

A technique facilitating an analysis without any a priori knowledge about the system under investigation is the method of Distribution of Relaxation Times (DRT). This method is further associated with an improved data resolution in frequency domain.^[2] Thus, due to its potential towards advanced data analysis, its application has gained popularity in recent

impedance research.^[3–9] Nevertheless, the calculation of DRT plots is an intrinsically ill-posed problem and regularization methods are needed.^[10] Therewith, new problems finding optimal DRT solutions come along. An often-used mathematical method is the Tikhonov regularization. Here, the central problem is to find a proper regularization parameter, which controls the extent of filtering noisy EIS data. In this work, a new approach is proposed to find optimal regularization parameters, which is based on a small number of repetitive impedance measurements in combination with a statistical resampling method. In addition, the new test function is combined with a current approach of this research field. Finally, the approaches are tested for simulated EIS spectra as well as for experimental datasets.

2. Theory

A key problem in interpreting impedance data is to find a meaningful model to describe the system under investigation. As a consequence, an equivalent electric circuit (EEC) representing the system's impedance behavior has to be constructed. After finding a suitable EEC, consisting of passive electrical elements, the model can be fitted to the measured data, leading to a quantification of experimental values. For this purpose, in 1977 MacDonald developed the Complex Nonlinear Least-Square (CNLS) approach for the fitting of impedance data to an appropriate EEC.^[11] This expansion of conventional nonlinear least-square problems takes the complex character of the impedance into account.

However, in general, the total number of physicochemical processes, determining the investigated system, is not known a priori. Therefore, ad hoc assumptions based on physical intuition have to be made. Moreover, discrete process time constants tend to overlap for most real electrochemical systems and the assignment of an eligible model can get ambiguous. As

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a proper EEC is crucial for interpreting impedance data, the latter problems can lead to vague results in EIS analysis. In this context, the method of Distribution of Relaxation Times is suitable to overcome these issues. Similar to the CNLS-fitting it yields a possibility to evaluate measured impedance data. Furthermore, a DRT analysis can provide the required information to build up EECs, which can be used in a following CNLS-fitting.^[12]

The DRT method deconvolutes the collected data from the frequency domain (ω) into the time domain (τ), leading to a direct access to the isolated time constants of the system's electrochemical processes.^[10] For an impedance spectrum, containing only polarization processes, the experimental values $Z_{\text{exp}}(\omega)$ and the distribution of relaxation times $g(\tau)$ are connected via equation (1).^[13]

$$Z_{\text{exp}}(\omega) = \int_0^{\infty} \frac{g(\tau)}{1 + i\omega\tau} d\tau \quad (1)$$

This equation (1) has the general form of a Fredholm integral equation of the first kind,

$$u(r) = \int_a^b K(r,s)f(s)ds \quad (2)$$

Here, the target distribution function $f(s)$ is associated with the experimentally measured data function $u(r)$ by the known kernel $K(r,s)$. Determining $f(s)$ from the experimental data is known to be ill-posed for such mathematical problems.^[14] Hence, any found solution may not be unique and is strongly prone to experimental instabilities.^[15,16] It can thus be shown that adding high frequency noise to the distribution function has almost no influence on function $u(r)$ as the integral in equation (2) operates as a smoothing filter. Conversely, small experimental errors yield strong fluctuations in the distribution function.^[14] Therefore, special mathematical techniques to avoid artifacts and erroneous oscillations have to be applied. References^[15,17–19] give an overview about the numerous approaches to determine the DRT from experimental impedance data. The in this work used approach is based on the Tikhonov regularization (also known as ridge regression),^[20,21]

$$\min_x F_{\lambda} = \underbrace{\|Ax - y\|^2}_{\text{least squares}} + \lambda \underbrace{\|x\|^2}_{\text{penalty}}, \quad (3)$$

in which x and y belong to two separable Hilbert spaces of finite or infinite dimensions which are connected via the operator A . In equation (3), a penalization term is added to a least-square method. Both, sum of squared residuals and penalty are minimized simultaneously. Therein, the regression error (first term) determines the fitting quality of the calculated model to the experimental data. On the other hand, the second term considers oscillations of the regression and smooths noise.^[15] To take given data's quality in account, the penalty is weighted by the regularization parameter λ . In particular, a high regularization parameter prevents data over-interpretation in the calculated solution (called overfitting), but at the same time it generates a bias between model and experiment. Thus, λ should be chosen as small as possible to avoid underfitting. The effect of different regularization parameter values is exemplarily shown for the simulated impedance spectrum in Figure 1.

The impact of different weighting of the penalty is obviously depicted in Figure 2. Here, DRT plots for divergent λ values are shown. The distribution function $g(\tau)$ is plotted against the time constant τ for each regularization parameter. All three plots were calculated from the same simulated impedance data, shown in Figure 1. Since in the Nyquist plot of Figure 1 two distinct semi circles are visible at least two DRT peaks can be expected. Besides, the model of the impedance simulation is well known to consist of two R-CPE circuits which are connected in series. Therefore, a regularization parameter of $\lambda = 10^0$ leads to underfitting in that case. Furthermore, as the equivalent circuit is well known in this case, it is clear that the regularization overestimates oscillations for $\lambda = 10^{-7}$. For this example, a regularization parameter value of 10^{-1} seems to yield outcome in good accordance to the known model. Apparently, the regularization parameter should be chosen as small as possible to guarantee good resolution in time domain and as large as required to avoid false peaks in the resulting DRT spectra.

Since a proper regularization parameter is crucial to generate valid DRT data, objective criteria are required to choose λ in connection with given experimental data.^[22] First approaches were presented by Saccoccio, et al.,^[15] who devel-

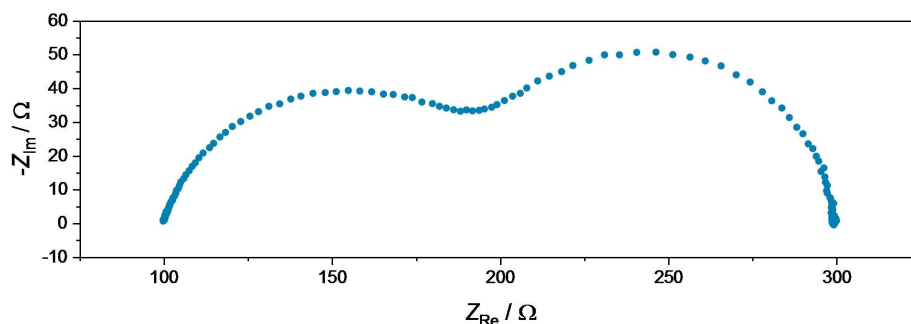


Figure 1. Simulated impedance spectrum. Simulated with two R-CPE circuits connected in series with an ohmic resistor. Where $R_0 = R_1 = R_2 = 100 \, \Omega$, $Q_1 = 10^{-5} \, F$, $Q_2 = 10^{-4} \, F$, $P_1 = 0.8$, $P_2 = 0.95$. Error factor $\varepsilon = 0.001$.

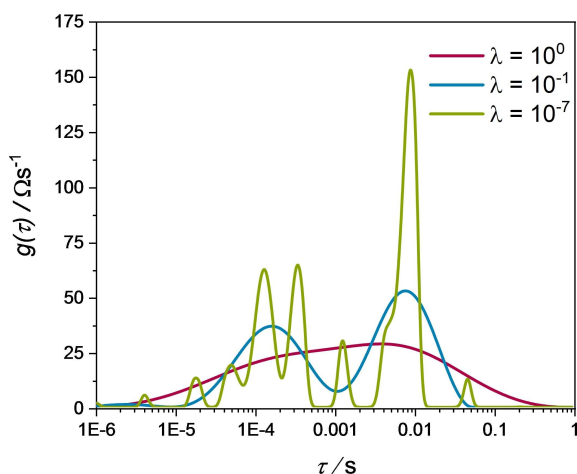


Figure 2. DRT plots for different regularization parameters $\lambda = 10^0$ (red), $\lambda = 10^{-1}$ (blue), $\lambda = 10^{-7}$ (green), calculated from impedance data in Figure 1.

opened two tests for finding the optimal λ value. The real and imaginary discrepancy test function $RID(\lambda)$ compares DRT spectra calculated for different regularization parameters from real part impedance data (DRT_{Re}) with those obtained from imaginary datasets (DRT_{Im}):

$$RID(\lambda) = \|DRT_{Re,\lambda} - DRT_{Im,\lambda}\|_2^2 \quad (4)$$

The second test is a cross-validation (CV). Herein, DRT spectra are calculated from real (imaginary) part impedance data as a function of λ . These DRT datasets are used to recalculate impedance spectra. Afterwards, their imaginary (real) datasets are matched with the imaginary (real) part of the experimental data to give the test result $CV'(\lambda)$ ($CV''(\lambda)$) in dependency of the regularization parameter (see equation 5).

$$\begin{aligned} CV'(\lambda) &= \|\mathbf{Z}'_{exp} - \mathbf{Z}'(DRT_{Im,\lambda})\|_2^2 \\ CV''(\lambda) &= \|\mathbf{Z}''_{exp} - \mathbf{Z}''(DRT_{Re,\lambda})\|_2^2 \end{aligned} \quad (5)$$

In other works, the Tikhonov regularization is understood in the context of Bayesian statistics, and the regularization parameter is additionally not treated as a scalar any longer but it is presented by a vector. By doing so, frequency dependent oscillations can be recognized in the regularization.^[23,24] In reference^[25] a new Tikhonov regularization without the usage of any adjusting parameters has been presented. In that approach a distribution of the regularization parameter across the time-scale of relaxation times was postulated. Using a special algorithm, the DRT solution was calculated simultaneously with the distribution of λ .

3. Results

3.1. Simulated Data

In order to determine the optimal size of the regularization parameter, we combined the above-described cross-validation with a new approach based on repetitive experimental impedance measurements and the subsequent calculation of variances.

A suitable λ optimally leads to DRT results free of oscillations. Therefore, the comparison of several DRT plots, gained from repetitive impedance measurements, should result in only small variances if the respective underlying impedance spectra only deviate in stochastic error and the filtering works well. Insufficient filtering (too small λ value), in contrast, is expected to provide diverging regularization outcome when pristine impedance data is superimposed by noise. In that case, this effect would be reflected by increasing variance of the resulting DRT plots.

Figure 3 shows two simulated Nyquist plots, which were generated from an electric circuit consisting of two R-CPE sub-circuits connected in series. For the calculation of both spectra, the same parameters $R_{0,1,2}$, $Q_{1,2}$ and $P_{1,2}$ were used (cf. Table 1). In addition, the simulated data is overlain with random noise of the same overall extent. Thereby, they only differ in random error structure and can, therefore, be interpreted as repetitive measurements. And yet, their deconvolution to time domain yields diverse outcomes.

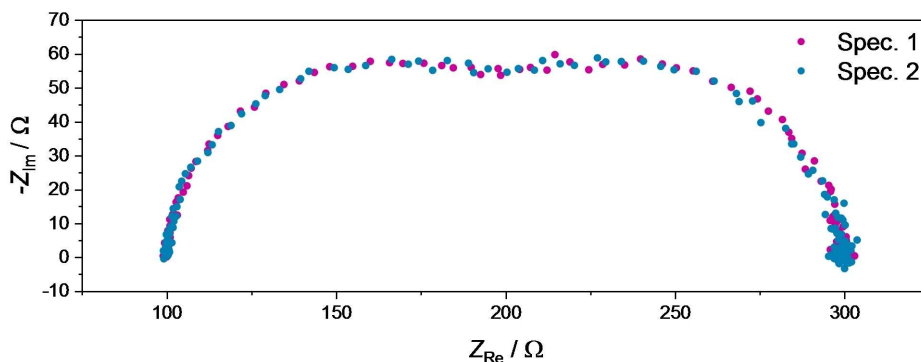
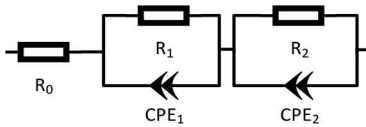
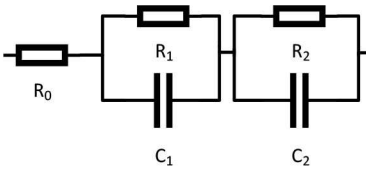


Figure 3. Simulated impedance spectra. Computed with two R-CPE circuits connected in series with an ohmic resistor. For detailed information see Table 1. Error factor is $\varepsilon = 0.5\%$. Both spectra were calculated with the same model but with diverging statistical error and can therefore be interpreted as repetitive measurements.

Table 1. Parameters of the simulated impedance spectra in Sections 2 and 3.1 and parametrization of the investigation of the electrical circuit.

Value	Unit	Simulated Data	Electrical Circuit
Electric circuit			
Chapter		2 and 3.1	3.2
Frequency range	Hz	10^5 – 10^{-3}	10^5 – 10^{-1}
Points per decade		20	30
ε	%	0.001, 0.01, 0.1, 0.5	/
R_0	Ω	100	10k
R_1	Ω	100	100k
Q_1	$s^{p_1} \Omega^{-1}$	$1 \cdot 10^{-5}$	$1 \cdot 10^{-8}$
P_1		0.95	1.00
R_2	Ω	100	100k
Q_2	$s^{p_2} \Omega^{-1}$	$1 \cdot 10^{-4}$	$2.02 \cdot 10^{-9}$
P_2		0.95	1.00

The effect of the generation of DRT spectra from such erroneous data is clearly depicted in Figure 4 (left side of the figure panel). Although both Nyquist plots contain the same information about the investigated system, the DRT results clearly differ when calculated with regularization parameters of 10^{-5} or 10^{-7} . On the other hand, the DRT peaks are in good accordance to each other for $\lambda = 10^{-3}$.

The deviation of the DRT outcome for a specific regularization parameter can be quantified by means of the variance $\sigma_{\tau_i}^2(\lambda)$. Thus, the variance of the N (here $N = 2$) DRT results $g_k(\tau_i, \lambda)$, which were generated with a specific λ value, has to be calculated for all N_{τ} time constants τ_i separately using equation (6),

$$\sigma_{\tau_i}^2(\lambda) = \frac{1}{N-1} \sum_{k=1}^N |g_k(\tau_i, \lambda) - \mu(\tau_i, \lambda)|^2. \quad (6)$$

The mean $\mu(\tau_i, \lambda)$ from equation (6) is defined by equation (7) as,

$$\mu(\tau_i, \lambda) = \frac{1}{N} \sum_{k=1}^N g_k(\tau_i, \lambda). \quad (7)$$

On right side of Figure 4, the respective outcome of this proceeding is shown for the DRT spectra shown on the left panel of the figure. Altogether, a trend is visible showing increasing overall variance values for decreased λ values. However, as the pristine impedance data is simulated two peaks have to result from the deconvolution to time domain. In consequence, the DRT spectra for the lowest filtering ($\lambda = 10^{-7}$) obviously reflect overfitting as both spectra contain additional false peaks of significant size. Additionally, for $\lambda = 10^{-5}$, spectrum 1 shows a distorted peak shape at lower time constants indicating a beginning overfitting. At the same time, these effects are connected with increasing variances. As a consequence, this correlation is used in a new approach for determining optimal regularization parameters.

In summary, DRT spectra were generated for several repetitive (simulated) impedance measurements using a defined set of regularization parameters. The deviation between the respective DRT plots as a function of the used regularization parameter was determined in the form of a variance. A sufficient filtering can then be assumed for the λ value that generates the lowest overall variance.

For a clearer representation, the variance as a function of the λ value $\sigma_{\tau_i}^2(\lambda)$ is summed up over all time constants τ_i using equation (8) (see Figure 5). As result, the total overall variance $\sigma_{\text{tot}}^2(\lambda)$ is obtained. The latter facilitates a simplified comparison of the effect of different regularization parameters and thus a minimum in variance is easy to determine.

$$\sigma_{\text{tot}}^2(\lambda) = \sum_i \sigma_{\tau_i}^2(\lambda) \quad (8)$$

In addition, a normalization is used in this work for the summarized variance to guarantee a better comparability between the results of different initial impedance spectra. For this purpose, the total variance depending on λ is divided by the total variance of the smallest regularization parameter $\lambda_{\min}$ since the latter is expected to show maximum σ_{tot}^2 values. Consequently, the highest total variance is equal 1 and equation (8) is transferred into,

$$\sigma_{\text{norm}}^2(\lambda) = \frac{\sum_i \sigma_{\tau_i}^2(\lambda)}{\sum_i \sigma_{\tau_i}^2(\lambda_{\min})}. \quad (9)$$

Figure 5 (a) presents a collocation of variances, which were calculated for the example of Figure 3 and Figure 4. Again, the variance was estimated as a function of the regularization parameter. The before seen trend is also visible in this graph as the total variance increases with decreasing λ values. Besides, this tendency is obviously depicted by the graph of Figure 5 (b). Here, the summarized and normed variance σ_{norm}^2 is plotted against λ yielding a sigmoidal curve. The improved clarity of

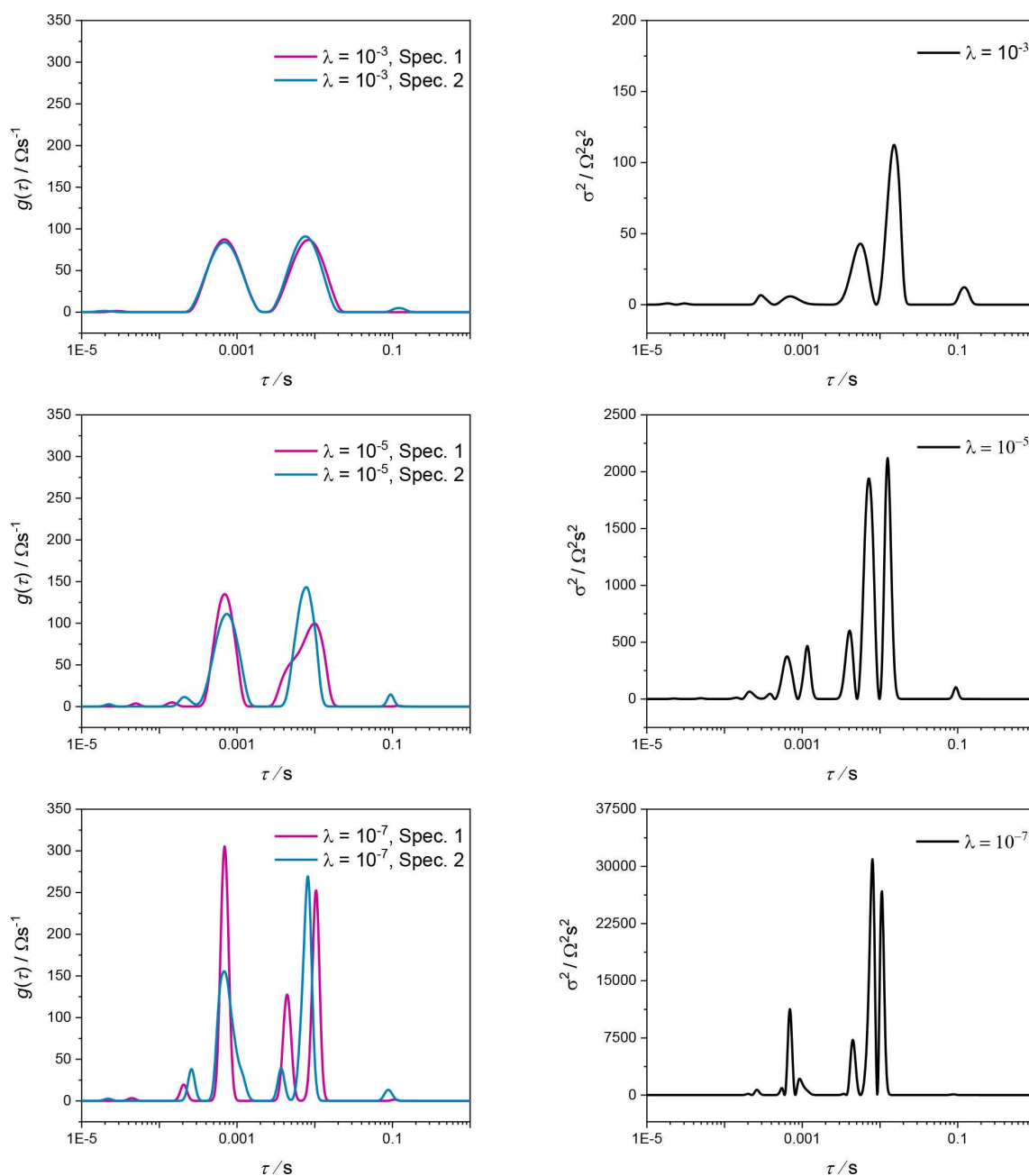


Figure 4. Left side: DRT spectra calculated with diverging regularization parameters ($\lambda = 10^{-3}$, $\lambda = 10^{-5}$ and $\lambda = 10^{-7}$) from the impedance spectra shown in Figure 3; Right side: variance of the respective DRT results in dependence of λ . The divergent scaling of the ordinate has to be considered.

this presentation form is evident and henceforth, only summarized variance values will be shown in this work.

The minimal regularization parameter in Figure 5 (b) ensuring a sufficient filtering is clearly identifiable to be $\lambda = 10^{-4}$, due to the fact that the total variance steeply rises for λ values below this threshold value.

Unfortunately, for the calculation of stable variance values as a function of λ often a vast amount of raw impedance data is necessary. Due to the time-consuming character of electrochemical impedance measurements the number of repetitive measurements should, on the other hand, be as small as possible. To overcome this situation, random data resampling is

a useful approach as it allows the generation of additional artificial datasets by resampling the initial available information ($\text{Re}_{\text{exp},k}$ and $\text{Im}_{\text{exp},k}$). The resampling is done for each real and imaginary value at the N respective frequencies ω_i for M times by loading the S measured datasets into Algorithm 1,

Algorithm 1: Resampling of the experimental impedance data.

Require: number of resamples M ; experimental datasets $\text{Re}_{\text{exp},k}(\omega_i)$ and $\text{Im}_{\text{exp},k}(\omega_i)$, $k = 1, \dots, S$ and $i = 1, \dots, N$

1. Evaluate $S \leftarrow$ number of experimental datasets

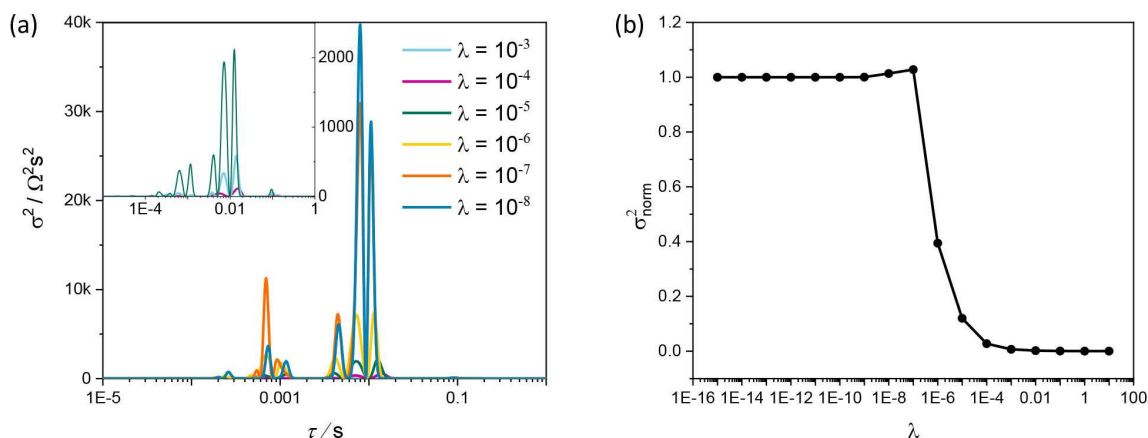


Figure 5. Variance of the DRT results of spectra 1 and 2 shown in Figure 3 in dependence of λ (a) and the summarized and normed variance $\sigma_{\text{norm}}^2(\lambda)$ plotted against the regularization parameter (b).

2. Evaluate $N \leftarrow$ number of elements per dataset
3. **for** $j = 1$ to M
4. **for** $i = 1$ to N
5. draw $k \sim \mathcal{U}\{1, S\}$
6. $\text{Re}_j(\omega_i) = \text{Re}_{\text{exp},k}(\omega_i)$
7. draw $k \sim \mathcal{U}\{1, S\}$
8. $\text{Im}_j(\omega_i) = \text{Im}_{\text{exp},k}(\omega_i)$
9. **end**
10. **end**

Afterwards, for each generated dataset, DRT plots dependent on λ are calculated and the results are compared by means of variance. Figure 6 exemplarily depicts the advantage of the data resampling. Here, three simulated datasets were used to calculate variance as a function of the regularization parameter. In Figure 6 (a), the two presented variance test results were each calculated based on two impedance data sets (spectrum 1 vs spectrum 2 and spectrum 1 vs spectrum 3). Although all spectra were calculated with the same equivalent circuit and they only differ in their errors, varying results can be obtained. As the resulting curves also show divergent optimal regulariza-

tion parameters, the incorporation of only two datasets seems to result in ambiguous information. Therefore, a 20fold resampling was used to generate the curves in Figure 6 (b). The data resampling obviously leads to more stable results, independent from the origin impedance data, as both curves overlap. Here, the dashed black line shows the course of the results for a variance test with 20 simulated spectra and without resampling. The latter perfectly fits the results which were calculated with the resampling. The examination of even two experimental datasets leads to stable test results if resampling is performed. Hence, a random resampling of the available experimental data should always be conducted, also because such calculations are easy to automate with an appropriate software tool.

Whereas the variance calculation shown above presents a minimum limit for the regularization, a limit towards too strong filtering seems to be accessible via the already mentioned cross-validation (CV) test, as can be seen in Figure 7. Here, a comparison of both test results and the respective DRT spectra is shown. Both test functions were calculated on basis of the impedance spectra from Figure 3, with the DRT spectra obtained for spectrum 1. In contrast to the variance test results,

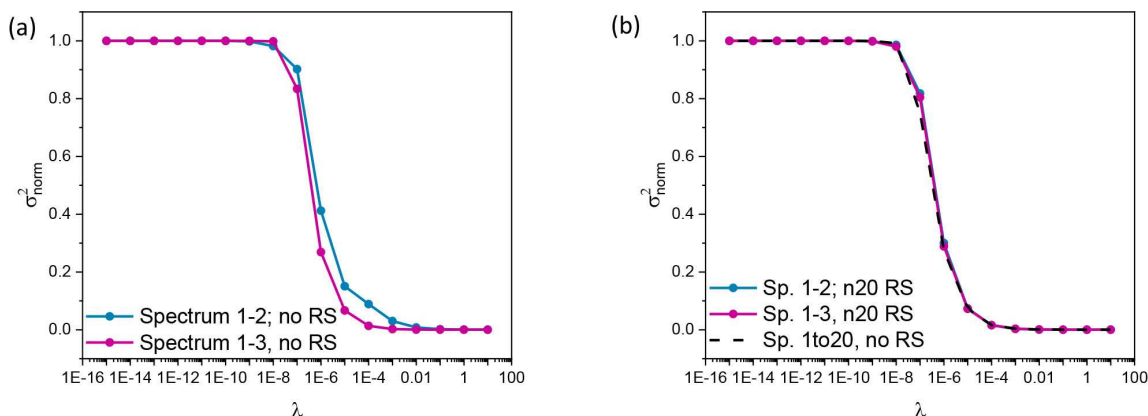


Figure 6. Effect of the resampling procedure. Normalized variance values calculated each for two repetitive impedance results (1 vs. 2 and 1 vs. 3) without (a) and with (b) twentyfold resampling. The results are calculated from simulated spectra with $\varepsilon = 0.5\%$ (cf. Table 1).

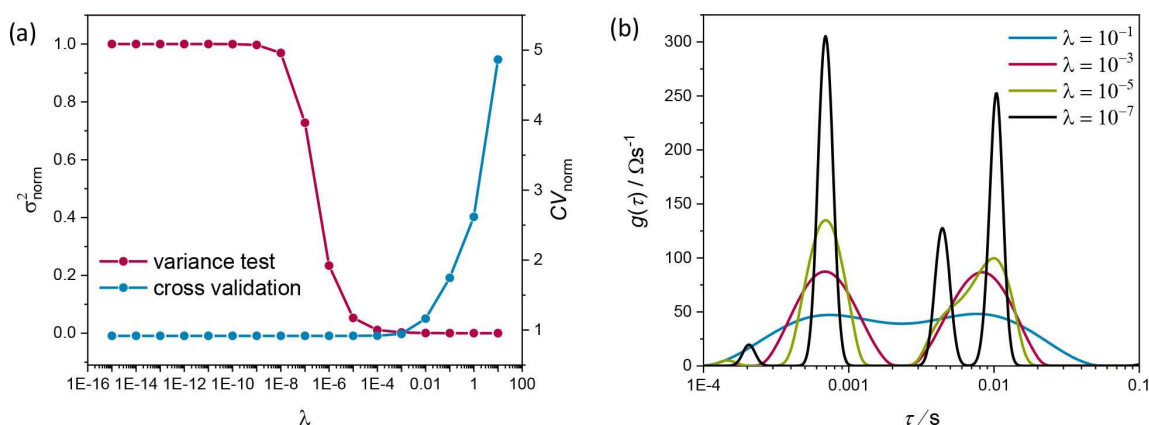


Figure 7. Combination of normalized variance test and cross-validation to find optimal λ (a). DRT plots calculated for varying λ values (b). All calculated from the impedance data of Figure 3.

the CV outcome increases for high regularization parameters. Again, a threshold value is identifiable for that the test results begin to increase progressively. In this case, increasing CV values indicate beginning underfitting, where information of the pristine dataset becomes neglected and plain DRT plots result. This can lead to overlapping peaks (see $\lambda = 10^{-1}$ in Figure 7 (b) for example). Thus, while insufficient filtering results in additional false peaks or peak distortion in the respective DRT plot (see $\lambda = 10^{-5}$ and $\lambda = 10^{-7}$ in Figure 7 (b)), underfitting oversimplifies the output by truncating single peaks. The optimal λ , therefore, can be found at minimal CV and minimum variance ($\lambda = 10^{-4}$ – $\lambda = 10^{-3}$).

To clarify the sensitivity of the herein presented variance test and cross-validation to the quality of a given impedance spectrum, simulated impedance data, superimposed with oscillation errors of different size, is processed via DRT method. For this purpose, a scaling factor ε was introduced that is multiplied with the simulated oscillation error (c.f. chapter 4). Four different ε values were used in this investigation (0.001, 0.01, 0.1 and 0.5%). For all datasets, DRT plots were calculated regarding regularization parameters between 10^{-15} and 10^1 (on a logarithmic scale). The received test results are presented in Figure 8 (a) and (b). Besides, selected DRT plots are shown in Figure 8 (c) and (d).

As expected, the variance curves shift on the abscissa to higher regularization parameters with increasing error, meaning that stronger filtering is required (see Figure 8 (a)). Furthermore, a rising need for filtering uncertain data can also be observed in the respective DRT plots (see Figure 8 (c)). Here, a splitting of the two main peaks in several new peaks of relevant size occurs even at tight filtering ($\lambda = 10^{-6}$) when higher oscillations ($\varepsilon = 0.5$) are added to the raw impedance data. At the same time, the qualitative shape of the curves stays unaltered. Therefore, it can be assumed that the variance test provides information about optimal regularization also for less erroneous impedance data.

On the other hand, mainly a slight shift on the ordinate, is visible for the cross-validation method (see Figure 8 (b)). This curve shift seems to be systematic. Thus, the plateaus of the

curves at low regularization parameters are shifted towards ascending CV values for increasing stochastic error in the pristine impedance data. Besides, the CV values tend to increase to a small degree for even lower regularization parameters when the superimposed error decreases. And yet, the dependency of the CV results from the error structure is not that distinct as it is for the variance test.

However, this result was expected as the CV results should only increase in the regularization range of underfitting, which is mainly affected by the distribution of the searched time constants itself. So, time constants that are close to each other are expected to yield underfitting even for lower regularization parameters and vice versa. Moreover, these findings are in good accordance to the underlying DRT results. According to Figure 8 (d) the two main peaks are baseline separated for regularization parameters below $\lambda = 10^{-2}$, regardless of error intensity ε . Only from higher λ values the peaks start to merge. In the case of high regularization parameters, the DRT spectra are further in good accordance to each other if calculated with the same λ value. The error structure's influence on the peak shapes is quite low, because filtering is distinct within the range of high regularization parameters. Minor deviations can only be found for the highest error (0.5%).

3.2. Experimental Measurements

The findings of the theoretical investigations were adapted to an experimental system. As the respective system has to be well known for a validation of the above methods, an electrical circuit was measured by means of impedance spectroscopy. The latter was composed by two R–C circuits, connected in series. To review the sensitivity of the cross-validation test and the variance test, the impedance measurements were conducted with different excitation amplitudes. Thereby the signal to noise ratio is varied and a different error extent is expected to be superimposed to the respective spectra. Overall, five different voltage amplitudes were used (2 mV, 4 mV, 6 mV, 8 mV and 10 mV). Figure 9 exemplarily shows the measured

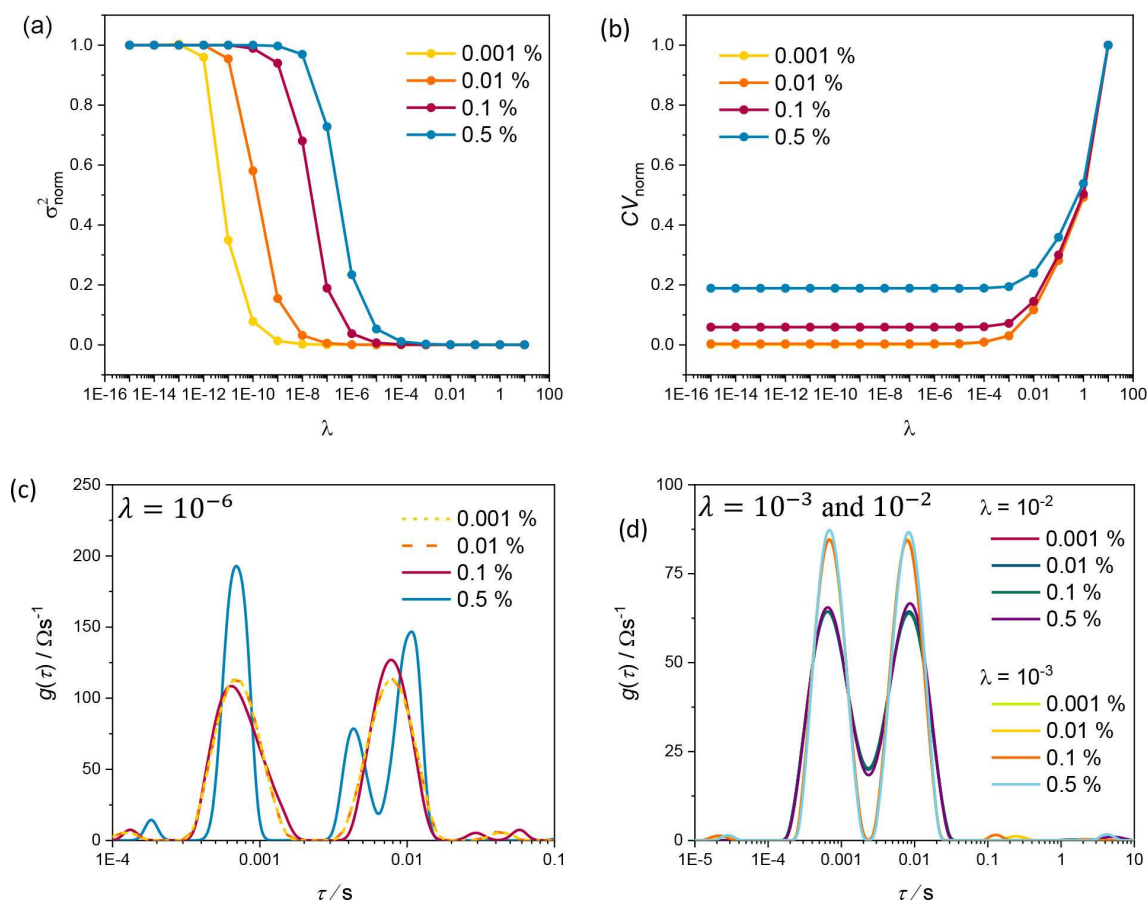


Figure 8. Totalized and normalized variance (a) and CV (b) depending on regularization parameter λ for the DRT deconvolution of the impedance spectrum from Figure 5 superimposed with increasing error factor ε . And respective DRT plots, calculated with $\lambda = 10^{-6}$ (c) and $\lambda = 10^{-3}$ and $\lambda = 10^{-2}$, respectively (d).

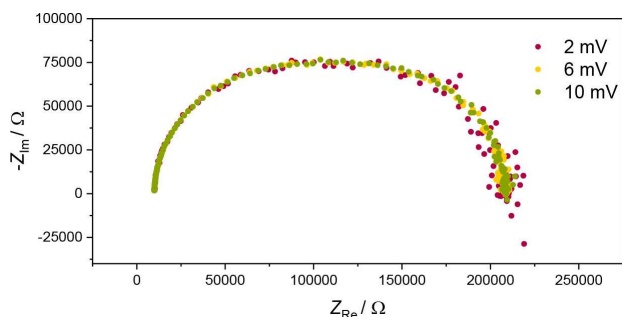


Figure 9. Impedance spectra of an electrical circuit consisting of two R-C circuits which were connected in series. The measurement was conducted with three different excitation amplitudes (2 mV – red, 6 mV – yellow, 10 mV – green) to gain impedance data of diverging quality.

impedance data for three voltages. It is obvious that the impedance data is the more erroneous the lower the excitation signal is. Besides, this phenomenon is more pronounced for lower frequencies than in the high frequency region. Hence, an impact from the input voltage on the DRT output of these measurements is expected. The results of a respective Kramers-Kronig analysis for the measured impedance data can be found in the supporting information.

First, both tests were reviewed in general by calculating DRT spectra for a selected voltage signal. Here, 8 mV was chosen as input amplitude. The measurement was conducted twice and both tests were executed. Figure 10 (a) shows a combination of both test results. Additionally, the respective DRT plots of one of the impedance measurements are shown in Figure 10 (b). Both test curves show the typical course that was already found for the simulated impedance data. Thus, for both tests threshold values can be determined indicating underfitting and overfitting respectively. Here, optimal regularization parameters lie in a range between 10^{-5} and 10^{-4} . For smaller values, the variance test results start to increase. On contrary, for higher λ s the CV results slightly increase. At the same time, the respective DRT plots show evidence of underfitting and overfitting outside that range. As a result, the two expected peaks start to overlap for $\lambda = 10^{-3}$ and merge for even higher regularization parameters to only one peak. On the other hand, a third peak becomes apparent for a regularization parameter of 10^{-7} . Eventually, optimal filtering is found for a regularization parameter of 10^{-5} as two baseline-separated peaks arise which were expected for the well-known system under investigation. Thus, this finding is in perfect accordance to the results of the variance test and the cross-validation.

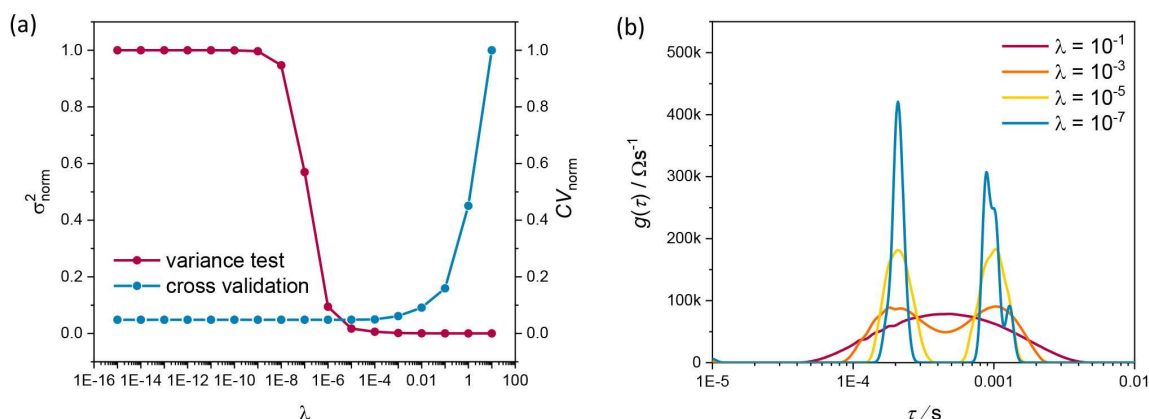


Figure 10. Combination of normalized variance test and normalized cross-validation to find optimal λ (a). DRT plots calculated for varying λ values (b). All calculated from impedance measurements of an electrical circuit (excitation amplitude = 8 mV).

The outcome for varying excitation amplitudes, and therefore different error extents, is shown in Figure 11. For the variance test, again a shift of the curves alongside the abscissa is visible. The higher the voltage the lower the regularization parameters that lead to proper DRT results. Notwithstanding, this effect is less distinct than for the simulated data of chapter 3.1. Here, the shifting seems to be regressive for rising amplitudes. This effect could be caused by a non-linear behavior between excitation voltage and resulting error size for instance.

Similarly, to the simulated data, the CV results, here, show a shift alongside the ordinate. The smaller the input voltage the more distinct the curve's displacement. As small excitation amplitudes implicate a worse signal to noise ratio, this finding is also in good accordance to the results of the synthetic experiments in chapter 3.1. Nevertheless, also a shift of the threshold value for an increasing CV can be found for varying excitation voltages. In this context, the dependency is even more pronounced as for the simulated impedance data. This could be due to the composition of the investigated electrical board. The single components were chosen in such a way that

the respective time constants are close to each other. For this measurement, the latter are even less separated than they are for the simulated data from chapter 3.1. Furthermore, capacitors were used instead of CPEs, as the latter does not exist as real electrical components.

In other words, the combination of the two tests gives a good evidence for finding optimal regularization parameters, also when real systems are investigated. In addition, the correlation between error structure and test result is the same as for the simulated data.

Experimental Section

Simulated and Measured Data

To avoid inverse crimes,^[26] all simulated data was generated in the frequency domain. The equivalent circuit that was used consisted of two parallel circuits connected in series, each with an ohmic resistor (R_1 and R_2) and a constant phase element (CPE₁ and CPE₂). In addition, an ohmic resistor (R_0) was connected in series (see Table 1). Conclusively, following equation (10) was used to calculate the simulated impedance spectra $Z_{\text{model}}(\omega)$,

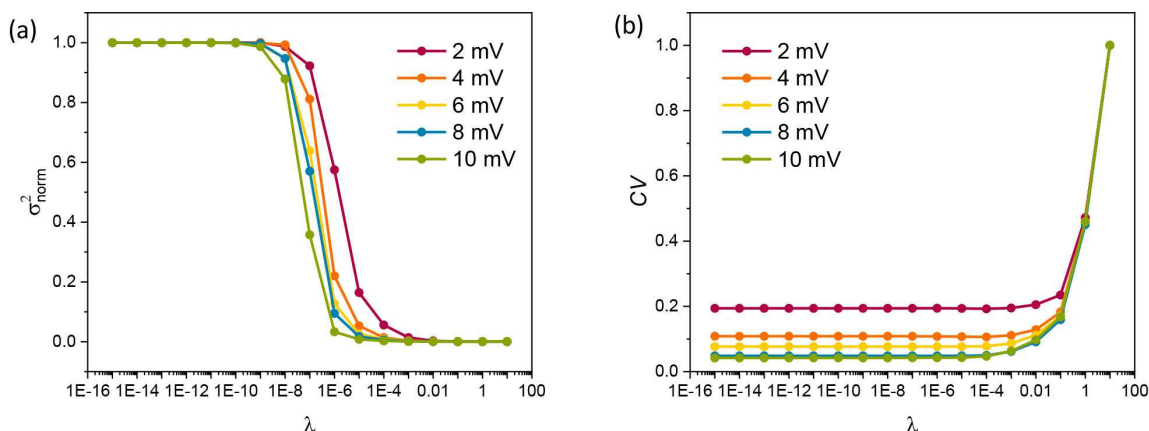


Figure 11. Totalized and normalized variance (a) and CV (b) depending on regularization parameter λ . The underlying DRT plots were calculated from impedance measurements of an electrical circuit. Varying input voltages were used.

$$Z_{\text{model}}(\omega_i) = R_0 + \left(\frac{1}{R_1} + Q_1(i \cdot \omega_i)^{P_1} \right)^{-1} + \left(\frac{1}{R_2} + Q_2(i \cdot \omega_i)^{P_2} \right)^{-1} \quad (10)$$

The CPEs are defined by two values, Q_i and P_i . The latter is an exponential factor and determines the aberrance of the CPE's behavior from a pure capacitor. When $P_i = 1$, the parameter Q_i is the capacitance of a capacitor. Table 1 depicts the parametrization of the simulated spectra.

To generate more realistic data, stochastic error was added separately to the real part and the imaginary part of the modeled impedance $Z_{\text{model}}(\omega_i)$.^[15]

$$\begin{aligned} Z'_{\text{sim}}(\omega_i) &= Z'_{\text{model}}(\omega_i) + \varepsilon \cdot |Z(\omega_i)| \cdot N, \quad N \sim \mathcal{N}(0, 1) \\ Z''_{\text{sim}}(\omega_i) &= Z''_{\text{model}}(\omega_i) + \varepsilon \cdot |Z(\omega_i)| \cdot N, \quad N \sim \mathcal{N}(0, 1). \end{aligned} \quad (11)$$

The error term in equation (11) consists of a scaling factor ε which is multiplied with the modulus $|Z(\omega_i)|$ of the respective frequency,^[27,28] ensuring the heteroscedastic structure of stochastic error in impedance measurements,^[27,29] and a normally distributed random number N_i ($\mu = 0$ and $\sigma^2 = 1$), which is different for each frequency and for the imaginary and the real part as well. This leads to a damped error structure which is randomly distributed around an EV of 0.^[30] The error was added separately to the imaginary and the real part, as well as for individual frequencies.

For the experimental measurements an electrical circuit was built up from two R–C circuits which were connected in series with an additional resistor (see Table 1). All resistors had a tolerance of $\pm 5\%$. For the experimental impedance measurements, a VMP-3 potentiostat (Bio-Logic, France) was used. The measurements were conducted at 20 °C. As excitation signal, sinusoidal voltages were used with amplitudes of variable magnitude (2 mV, 4 mV, 6 mV, 8 mV and 10 mV). 30 points per decade were recorded in a frequency range from 100 kHz down to 100 mHz. Each frequency was averaged over two measurements.

Data Evaluation

All DRT plots were calculated using an extended version of the MATLAB application DRTtools, developed and introduced by Wan et al.^[10,31] Our extension incorporates the automated calculation of DRT plots with several regularization parameters. Additionally, the test functions described above (variance test and cross-validation test) were implemented. Furthermore, a software solution for the calculation of the simulated impedance spectra mentioned before was added to the GUI.

The DRT plots were calculated from the imaginary part of the simulated and measured impedance data. Here, we assume the validity of the Kramers-Kronig relations.^[32] For the discretization, Gaussian radial basis functions were used. The considered range of time constants was obtained from the reciprocal impedance data's frequency range added by one order of magnitude at lower and upper boarder. Afterwards, the number of time constants was calculated by multiplying the amount of measured/simulated frequencies with a factor of 10.

The standard procedure for the calculation of the variance test sum was based on two simulated spectra or two measured spectra respectively, which were resampled to 20 new artificial spectra. These were also the basis for the CV. In that case, the synthetic impedance spectra were averaged and eventually, the CV was

calculated for the mean impedance data. Furthermore, results of the variance test and the cross validation were scaled for better comparability. For the variance test, the test value at $\lambda = 10^{-15}$ was set as base and all remaining data was divided by the latter. Cross validation results were scaled by the test value at $\lambda = 10^1$.

4. Conclusions

In this work, a new approach for finding the best regularization parameters for generating DRT spectra was proposed. Repetitive impedance measurements and the comparison of the respective DRT plots offer an access for finding the minimum required size of the regularization parameter. To keep the experimental data acquisition as least elaborate as possible, a software-based resampling of the measured data was introduced, which allows the generation of synthetic datasets. Thus, measuring a certain impedance spectrum twice gave enough information for stable test values. Additionally, the cross-validation introduced by Saccoccio, et al. was used to determine an upper boundary for the regularization parameter. Both methods in combination showed well applicability for simulated as well as for experimental data. Here, the comparison of the test results with the respective DRT plots was in good accordance with the expectations, which were based on the well-known information used for the impedance simulation and measurement. Therefore, the combination of cross-validation and new variance test offers a superior possibility to find optimal regularization parameters in impedance deconvolution.

Naturally, this procedure is accompanied with additional impedance measurements yielding in increased time expenses. For this reason, it is advisable to use these tests only for initial impedance spectra if the remaining spectra are investigated for the same system and under comparable conditions.

In this work, in addition to the synthetic electrochemical impedance data, only an electrical system was investigated using the new test procedures. The application of the herein shown proceeding for finding optimal regularization parameters is still pending for real electrochemical impedance data. Therefore, a stable and well-known electrochemical system has to be examined by means of the new variance test and the cross-validation test in future work. Thus, the test results can be compared with the expectations for the respective measurements and finally the practicability of the combination of both tests for finding optimal regularization parameters can be validated for experimental electrochemical spectra.

Author Contributions Statement

All authors listed made a direct and intellectual contribution to the work and approved it for publication.

Conflict of Interest Statement

The authors exclude any personal, professional or financial relationships that could potentially be construed as a conflict of interest.

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